Assignment 3: Keyboard Layout Optimization

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1 Introduction

The aim of this assignment is to visualize how keyboard layout affects typing effort and to use simulated annealing to optimize the layout for a given text.

2 Method

- Start from QWERTY layout coordinates.
- Cost function: sum of Euclidean distances between consecutive characters.
- Optimization step: swap two randomly chosen characters.
- Simulated annealing:
  - Always accept better layouts.
  - Accept worse layouts with probability $e^{-\Delta/T}$. Δ is the change in the cost function.
  - Temperature decays geometrically ($T \leftarrow T \cdot \alpha$).

3 Results

3.1 Optimization Trace

Figure 1 below shows the best cost and current cost over iterations. The best cost decreases steadily, with occasional increases in the current cost due to probabilistic acceptance of worse solutions.

3.2 Final Layout

Figure 2 below shows the optimized keyboard layout after simulated annealing.

3.3 JSON Layout

A snippet of the final layout saved in `final_layout.json` is shown below:

```
{
  "q": [7.0,2.0],
  "w": [8.5,1.0],
  "e": [4.5,1.0],
  "r": [5.0,2.0],
  ....
}
```

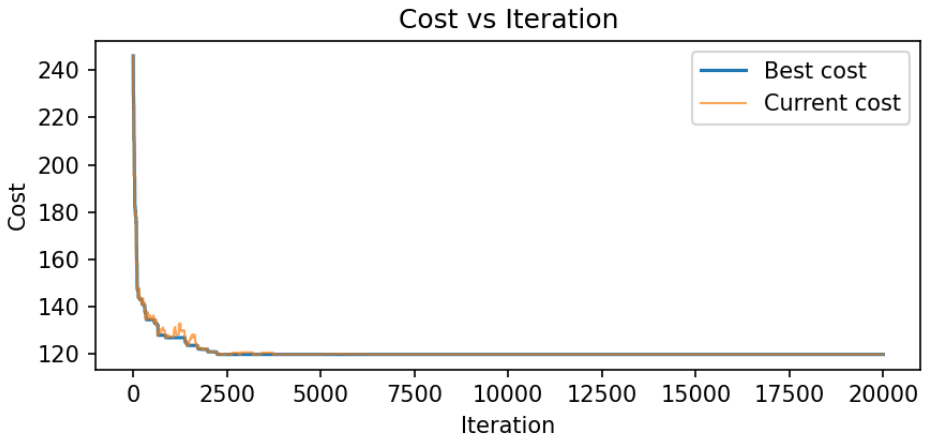


Figure 1: Cost vs Iteration during simulated annealing.

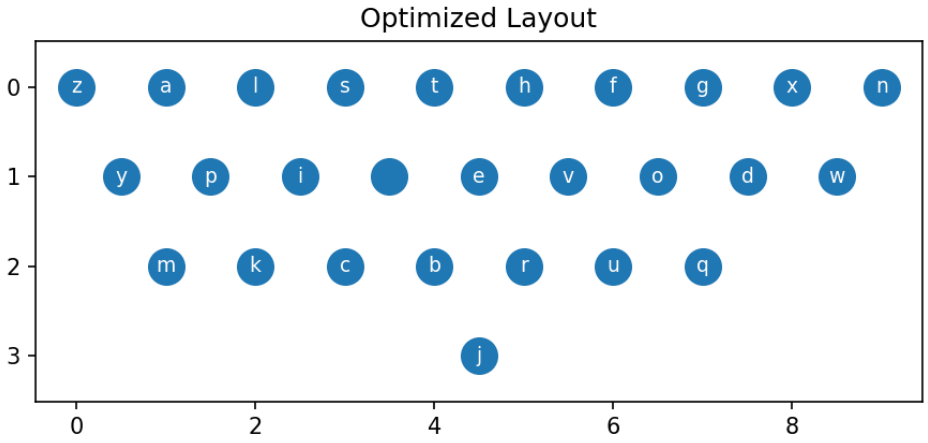


Figure 2: Final optimized keyboard layout.

The `final_layout.json` has also been submitted in the .zip file along with others.

4 Effect of Parameters

- **Iterations (iters):** Increasing iterations improves the final cost and smooths convergence. Too few iterations can freeze early at suboptimal layouts.
- **Initial temperature (t0):** Higher $t0$ allows more exploration early on. Too high leads to random wandering; too low gets stuck quickly.
- **Decay factor (alpha):** Values close to 1.0 mean slow cooling (better results, longer runtime). Smaller values mean faster freezing (less exploration).

5 Instructions to Run

```
# Run with sample text

python3 kbd_optim.py --textfile assignments/kbd_optim/sample_text.txt
```

```
# Run with a custom dataset

python3 kbd_optim.py --textfile bigtext.txt

# Change annealing parameters (edit in code)

params = SAParams(iters=50000, t0=2.0, alpha=0.999)
```

6 Conclusion

The simulated annealing optimizer greatly lowers typing effort compared to the standard QWERTY layout. The effectiveness of the optimized layout depends heavily on the choice of annealing parameters, which control the balance between exploring new possibilities and refining existing solutions.